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Indian Institute of Technology Bhilai
Department of Computer Science and Engineering
Database Management Systems
(CSL303/MAL505/CS254)

Mid-Semester, Date: September 24, 2025

Timing: 09:30 AM to 11:30 AM

2025-26-M Semester

Max mark: 50

Consider the scenario of a courier company trying to build a record keeping system for its operation. The following are the requirements stated about the operation of the company

- Each consignment has an alphanumeric unique consignment id, sender's name, sender's address, sender's phone, receiver's name, receiver's address, receiver's phone and cost paid by the sender. ✓
- The courier company has few warehouses throughout the country. The warehouses have a unique 5 character alphabet code along with their name, city and pin ✓
- The event of the item entering a warehouse and leaving a warehouse has to be recorded along with warehouse info, date-time of the event and a remark. ✓
- Each item goes through a list of warehouses before it reaches the last warehouse at the receiver's location. The list of warehouses that an item goes through does not contain any repetitions. •
- A item can have a special event called "Out for Delivery" when it exits the nearest warehouse of receiver's location. This event should be recorded similar to other events with date-time and remark.
- Each warehouse is connected to a few other warehouses. For example, if warehouse1 is connected to warehouse2 and warehouse3, all the items leaving warehouse1 can have only the following events next
 - Enter Warehouse2
 - Enter Warehouse3
 - Out for delivery
- An item can also have events called "Picked Up" and "Delivered". These events are also to be recorded in the same manner except that there are no warehouse information associated with these events.

1. Draw the Entity Relationship diagram for the database that would store the data for the courier company. (5)
2. Derive the relations and its schema from the ER Diagram. (4)
3. List the functional dependencies for the relational schema that needs to be satisfied. (5)

- Mongo* →
4. Perform Normalization and show the list of relational schema in 3NF and in BCNF. Specify the foreign key relationships as well. Write your steps explicitly. (6)
 5. Write the necessary DDL query in SQL for creating the relational schema with the obtained constraints. (4)
 6. Write the collections that you will create if you need to store the data of the courier company in a NoSQL database. Give the document structure for each of the collection with example. (6)
 7. Write SQL query AND MongoDB query to get the following results from the relational schema or Collections (respectively) produced.
 - (a) Given a consignment ID show the tracking details of the item. Show the city of the warehouses it is/was in, upto the current status along with date-time and remark. (4)
 - (b) Find the list of warehouses in decreasing order of the number of items processed by them. (4)
 - (c) Find the total number of items processed by the courier company as a whole with two given dates. The items still undelivered must also be counted. (4)
 - (d) Find the number of items that are currently in possession of the courier company (i.e. note yet delivered) (4)
 - (e) Find out the number of warehouses each city has. Do not consider the cities which does not have any warehouses. (4)
 8. Find all items which exits some warehouse1 and enters another warehouse2 next, where warehouse1 and warehouse2 are not connected. (5 (bonus))